

Engineering Statistics

Yili Hong

Chapter 8: Inference for Means

Outline

1. Student's t -distribution

- ▶ Why use the t -distribution when σ is unknown?

2. One-sample t -inference

- ▶ Confidence intervals and hypothesis tests for a population mean.

3. Paired t -test (Self-Study)

- ▶ Inference based on within-pair differences.

4. Two-sample t -inference

- ▶ Comparing the means of two independent populations.

Assumptions for inference about μ

1. We have a good **simple random sample** (SRS) from the population of interest.
2. The population is **Normally distributed** $N(\mu, \sigma^2)$.
3. **We don't know** both μ , and σ .

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

With unknown σ

- ▶ The sample standard deviation s provides an estimate of the population standard deviation σ .

$$S = \sqrt{S^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}.$$

- ▶ Standard error: $\frac{s}{\sqrt{n}}$.

t -distributed test statistic

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2),$$

► σ known:

$$T = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

► σ unknown:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1},$$

where t_{n-1} is the t -distribution with $n - 1$ degrees of freedom.

t —distributions

Let $Z \sim N(0, 1)$ and $U \sim \chi_k^2$. Assume that Z and U are independent. Let a random variable T be

$$T = \frac{Z}{\sqrt{U/k}}.$$

Then, the random variable T follows t —distribution, denote

$$T \sim t_k,$$

where k is the degrees of freedom.

t -distributions

It is known that

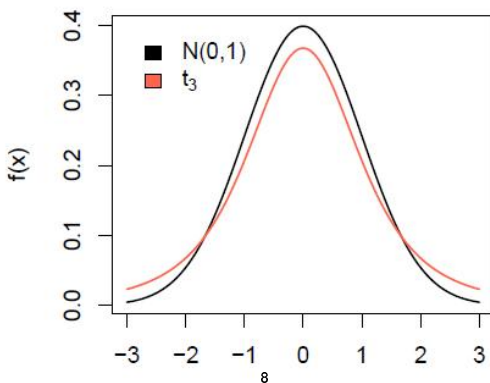
$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

and $\bar{X}$, S^2 are independent. Therefore,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

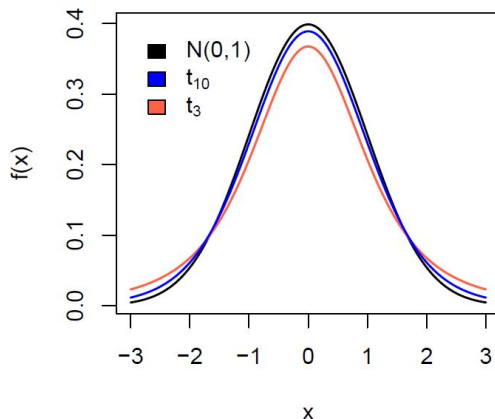
t -distributions

- ▶ The density curve of the t -distribution is similar to $N(0, 1)$ in shape.
 - ▶ Symmetric about 0, unimodal, and bell-shaped.
- ▶ The spread of t -distributions is a bit greater than that of $N(0, 1)$.
 - ▶ t -distributions have heavier tails.



With large sample size

- ▶ As sample size n increases:
 - ▶ The t density curve approaches the $N(0, 1)$ curve.
 - ▶ $S^2 \rightarrow \sigma^2$ by the law of large numbers.
 - ▶ $S \rightarrow \sigma$.



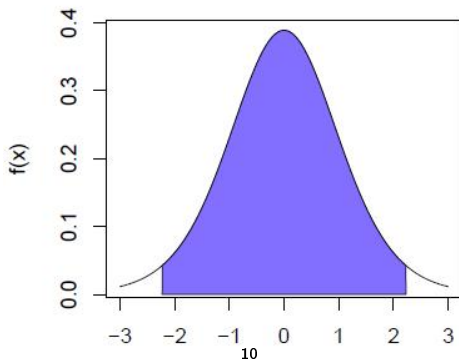
CI for a Normal population mean μ (σ unknown)

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

- ▶ μ is unknown $\Rightarrow$ our interest.
- ▶ σ is **unknown**.

A level $(1 - \alpha)100\%$ **confidence interval** for the population mean μ :

$$\bar{x} \pm t_{(1-\frac{\alpha}{2}, n-1)} \times \frac{s}{\sqrt{n}}$$



CI for a Normal population mean μ (σ unknown)

$$\bar{x} \pm t_{(1-\frac{\alpha}{2}, n-1)} \times \frac{s}{\sqrt{n}}$$

- ▶ $\bar{x}$: **estimate** (statistic)
- ▶ $t_{(1-\frac{\alpha}{2}, n-1)} \times \frac{s}{\sqrt{n}}$: the **margin of error**
 - ▶ $t_{(1-\frac{\alpha}{2}, n-1)}$ can be found by
 - ▶ R function: `qt()`
 - ▶ *t*-table

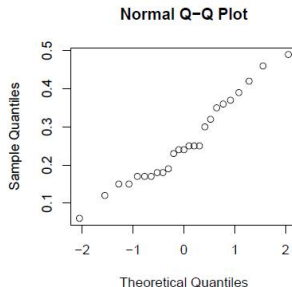
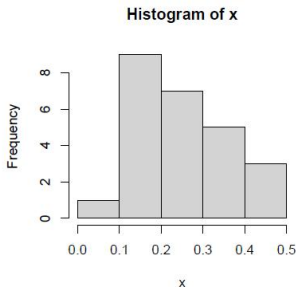
Example: Testosterone levels in adolescent males

- ▶ A study of testosterone levels in 25 males between the ages 14 and 20 years.
- ▶ Assumption:

$$x_1, \dots, x_{25} \stackrel{iid}{\sim} N(\mu, \sigma^2).$$

Example: Testosterone levels in adolescent males

- ▶ Check Normality assumption:



- ▶ Shapiro-Wilk test of normality
 - ▶ $H_0 : x_1, \dots, x_n$ came from a normal distribution.
 - ▶ P-value = 0.337

Example: Testosterone levels in adolescent males

- ▶ $\bar{x} = (x_1 + \dots + x_{25})/25 = 0.258$.
- ▶ $\frac{s}{\sqrt{n}} = \frac{0.112}{\sqrt{25}} \approx 0.022$
- ▶ With confidence level 95%, $t_{1-\alpha/2, n-1} = 2.06$

A 95% **confidence interval** for the population mean μ :

$$0.258 \pm 2.06 \times 0.022 \iff (0.213, 0.303).$$

One-sample t -test

- ▶ A hypothesis test for a population mean μ (σ unknown)
 - ▶ To test $H_0 : \mu = \mu_0$.
- ▶ Under $H_0 : \mu = \mu_0$ is true, the sampling distribution of the sample mean is

$$\bar{X} \stackrel{H_0}{\sim} N\left(\mu_0, \frac{\sigma^2}{n}\right).$$

- ▶ Then, the test statistic is

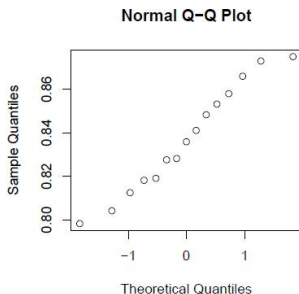
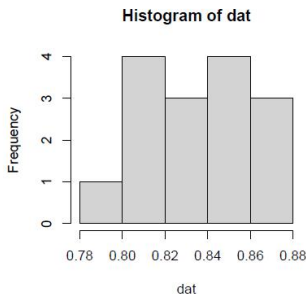
$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \stackrel{H_0}{\sim} t_{n-1}.$$

Example: Golf Club Design (Example 9-6)

- ▶ The coefficients of restitution (COR):
 - ▶ it measures the energy transfer between the golf clubface and the golf ball.
 - ▶ so it determines the performance and effectiveness of clubs.
- ▶ 15 drivers produced by a particular club maker were selected at random and their COR measured.
- ▶ Our interest is to determine if there is evidence to support a claim that the mean COR exceeds 0.82.

Example: Golf Club Design

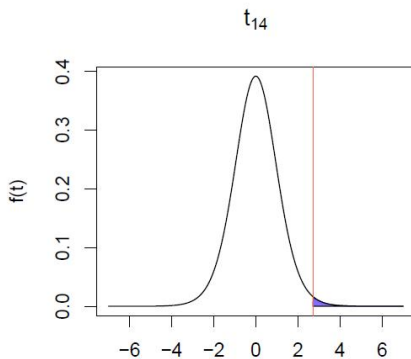
- ▶ $H_0 : \mu = 0.82$ vs $H_a : \mu > 0.82$
- ▶ Under $H_0 : \mu = 0.82$ is true, $X_1, X_2, \dots, X_{15} \stackrel{iid}{\sim} N(0.82, \sigma^2)$.
- ▶ Check Normality assumption:



- ▶ Shapiro-Wilk test
 - ▶ P-value = 0.7075

Example: Golf Club Design

- ▶ $\bar{x} = (x_1 + \cdots + x_{15})/15 = 0.83724$.
- ▶ $\frac{s}{\sqrt{n}} = \frac{0.0246}{\sqrt{15}} \approx 0.00634$
- ▶ The observed $t = \frac{0.83724 - 0.82}{0.00634} = 2.719$.
- ▶ P-value = $P(T > 2.719) = 0.0083$



Using AI: Understanding the t Procedure

Ask AI

Explain why the Student t distribution is used for inference about a population mean when σ is unknown. How does the sample size affect the degrees of freedom?

Verify

Check the role of the estimated standard error, normality assumptions, and the difference between the t and standard Normal distributions.

Matched pairs t tests

Sometimes we want to compare treatments or conditions at the individual level. The individuals in one sample are related to those in the other sample (**not independent**).

- ▶ Pre-test and post-test studies look at data collected on the same sample elements before and after some experiment is performed.
- ▶ Twin studies often try to sort out the influence of genetic factors by comparing a variable between sets of twins.

Matched pairs t tests

- ▶ To compare the responses to in a matched pairs design
 - ▶ Find the **difference** between the responses within each pair.
- ▶ The parameter μ_D is the **mean difference** in the responses within matched pairs of subjects in the entire population.
 - ▶ $H_0 : \mu_D = 0$
 - ▶ Apply the **one-sample t -test** to the differences.

Example: Parallel Park Cars (Example 10-11)

14 subjects were asked to parallel park two cars having very different wheel bases and turning radii. The time in seconds for each subject was recorded.

Subject	Automobile		Difference
	$1(x_{1j})$	$2(x_{2j})$	(d_j)
1	37.0	17.8	19.2
2	25.8	20.2	5.6
3	16.2	16.8	-0.6
4	24.2	41.4	-17.2
5	22.0	21.4	0.6
6	33.4	38.4	-5.0
7	23.8	16.8	7.0
8	58.2	32.2	26.0
9	33.6	27.8	5.8
10	24.4	23.2	1.2
11	23.4	29.6	-6.2
12	21.2	20.6	0.6
13	36.2	32.2	4.0
14	29.8	53.8	-24.0

Example: Parallel Park Cars

Q. Do the two cars have different mean parking times?

- ▶ $H_0 : \mu_D = 0$ vs $H_a : \mu_D \neq 0$
- ▶ For $i = 1, 2, \dots, 14$, X_{Di} = the difference between i th pair.
- ▶ Under $H_0 : \mu = 0$ is true, $X_{D1}, X_{D2}, \dots, X_{D14} \stackrel{iid}{\sim} N(0, \sigma^2)$.
- ▶ $\bar{x}_D = (x_{D1} + \dots + x_{D14})/14 = 1.21$.
- ▶ $\frac{s_D}{\sqrt{n}} = \frac{12.68}{\sqrt{14}} \approx 3.39$
- ▶ The observed $t = \frac{1.21-0}{3.39} = 0.357$.
- ▶ P-value = $2 \times P(T > 0.357) = 0.7268$.

Self-Study Can You Learn This Method with AI?

The Problem

Suppose we measure the same subjects before and after a treatment. How can we determine whether the treatment changes the population mean?

Key Idea: Analyze Within-Pair Differences

$$D_i = X_{i,\text{after}} - X_{i,\text{before}}$$

A paired t -test is a **one-sample t -test** on the differences.

Your AI Learning Task

Ask AI to explain the assumptions, confidence interval, and hypothesis test for paired data. Then verify the explanation using what you learned about the one-sample t -test.

Comparing two population means

Goal:

- ▶ To compare the responses to two treatments
- ▶ To compare the characteristics of two populations.
- ▶ It is important to determine if the two samples are **independent** or not.

Comparing two population means

Matched pairs samples:

- ▶ The individuals in both samples are **related**
 - ▶ e.g. repeated measures of the same subjects, sibilings, twins,
...
 - ▶ Matched pairs t -test

Independent samples:

- ▶ The individuals in both samples are chosen separately
(**independently**)
 - ▶ Two-sample t -test

Assumptions for inference of two population means

1. We have a good **simple random samples** (SRS) from the populations of interest.
2. The populations are **Normally distributed**: $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$.
3. **We don't know** $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$.

Notations

We can rewrite the **assumptions** as

$$X_{11}, X_{12}, \dots, X_{1n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2),$$

$$X_{21}, X_{22}, \dots, X_{2n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2).$$

Notations

Population	Variable	Population mean	Population Standard deviation
1	X_1	μ_1	σ_1
2	X_2	μ_2	σ_2

We want to compare the two population means, by giving a confidence interval for their difference ($\mu_1 - \mu_2$) or by the hypothesis testing of no difference. We will use $\bar{X}_1 - \bar{X}_2$.

Population	Sample size	Sample mean	Sample Standard deviation
1	n_1	$\bar{X}_1$	S_1
2	n_2	$\bar{X}_2$	S_2

t — distribution for two independent samples

- Assumptions:

$$X_{11}, X_{12}, \dots, X_{1n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2),$$

$$X_{21}, X_{22}, \dots, X_{2n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2).$$

- Two sample are **independent** and **Normally** distributed, so

$$(\bar{X}_1 - \bar{X}_2) \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

t — distribution for two independent samples

Then, approximately,

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_{df},$$

with the degrees of freedom (df):

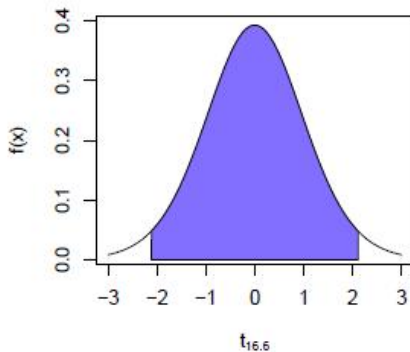
$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1} \left(\frac{s_2^2}{n_2}\right)^2}.$$

The approximation is accurate when $n_1 \geq 5$, and $n_2 \geq 5$.

Confidence intervals for $\mu_1 - \mu_2$

A level $(1 - \alpha)100\%$ **confidence interval** for $\mu_1 - \mu_2$:

$$\bar{x}_1 - \bar{x}_2 \pm t_{(1-\frac{\alpha}{2}, df)} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$



Example: Transgenic chickens

- ▶ Researchers created transgenic chickens resistant to avian flu infection.
- ▶ They compared the hatching weights (g) of 45 transgenic chickens and 54 commercial chickens of the same breed.
- ▶ Assumption:

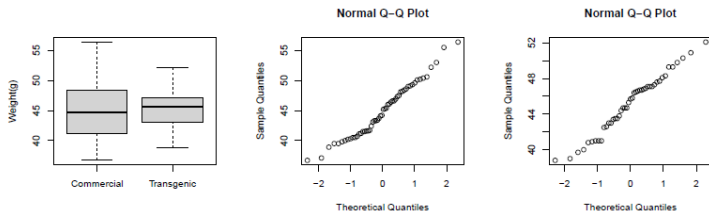
$$x_{11}, \dots, x_{1n_1} \stackrel{iid}{\sim} N(\mu_1, \sigma_1^2),$$

$$x_{21}, \dots, x_{2n_2} \stackrel{iid}{\sim} N(\mu_2, \sigma_2^2).$$

where $n_1 = 45$, $n_2 = 54$.

Example: Transgenic chickens

► Check assumptions:



► Shapiro-Wilk tests

- Transgenic chickens: P-value = 0.4626
- Commercial chickens: P-value = 0.2045

Example: Transgenic chickens

Grp	Condition	n	$\bar{x}$	s
1	Transgenic	45	45.14	3.32
2	Commercial	54	44.99	4.57

► $\bar{x}_1 - \bar{x}_2 = 0.15.$

► $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \approx 0.795$

► $df = 95.3$

► With confidence level 95%, $t_{1-\alpha/2, df} = 1.985$

A 95% **confidence interval** for $\mu_1 - \mu_2$:

$$0.15 \pm 1.985 \times 0.795 \iff (-1.428, 1.728).$$

Two-sample t -test

- ▶ We want to test whether there is any difference between two population means
 - ▶ $H_0 : \mu_1 - \mu_2 = 0$.
- ▶ Under $H_0 : \mu_1 = \mu_2$ is true,

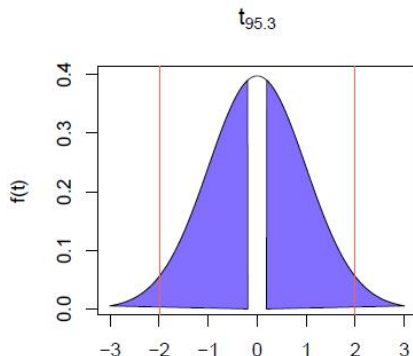
$$\bar{X}_1 - \bar{X}_2 \stackrel{H_0}{\sim} N\left(0, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

- ▶ Then, the test statistic is

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \stackrel{H_0}{\sim} t_{df}.$$

Example: Transgenic chickens

- ▶ $H_0 : \mu_1 - \mu_2 = 0$ vs $H_a : \mu_1 - \mu_2 \neq 0$.
- ▶ Under H_0 is true, $\bar{X}_1 - \bar{X}_2 \stackrel{H_0}{\sim} N\left(0, \frac{\sigma_1^2}{45} + \frac{\sigma_2^2}{54}\right)$.
- ▶ The observed $t = \frac{0.15-0}{0.795} = 0.19$.
- ▶ P-value = $2 \times P(T > |0.19|) = 0.8497$



Avoid the pooled two-sample t —procedures

There is another version of the two-sample t —methods:

- ▶ Further assumption: **equal variance (“pooled”)**
- ▶ The pooled (equal variance) two-sample t —test has degrees of freedom $n_1 + n_2 - 2$.
- ▶ However, the assumption of equal variance is **hard to check**.
- ▶ The two-sample t —methods with **unequal variances** assumption is **much safer**.

Using AI: Choosing a Two-Sample Method

Ask AI

A study compares two measurements from each subject. Should we use a paired t test or an independent two-sample t test? Explain the reasoning and how to check assumptions.

Verify

Confirm whether observations are matched, which differences are analyzed, and whether the proposed R code matches the study design.

Chapter Summary

- ▶ **Use the t distribution when population variability is unknown.** Estimate the standard error from sample data.
- ▶ **Match the method to the design.** Analyze within-pair differences for paired samples and compare group means for independent samples.
- ▶ **Use intervals and tests to quantify uncertainty.** Interpret results in terms of the population question, not just statistical significance.
- ▶ **Check assumptions and verify computation.**